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Project Proposal

(SCOPE DOCUMENT)

for

WZH Moments

Event Management Website

By

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Abstract

WZH Moments is an all-in-one event management platform developed using the MERN stack, designed to transform how events are organized, tracked, and overseen. It allows users to plan, manage, and follow events in real time, offering strong admin capabilities, insightful analytics, and effective communication tools for both organizers and attendees. One standout feature is its live tracking functionality, which lets users and administrators keep tabs on the event's progress as it happens enhancing transparency and overall engagement. With its scalable, user-friendly interface, WZH Moments tackles key event management challenges and introduces a smarter, more efficient way to handle events digitally.

Project Category

- Web Development
- Full Stack Application
- Real-time Systems

1. Introduction

The event management landscape is quickly changing with the rise of digital platforms. Traditional planning methods often struggle with manual coordination, lack of real-time tracking, and communication breakdowns. **WZH Moments** aims to solve these challenges by offering a unified, real-time event management solution built on the MERN stack (MongoDB, Express.js, React.js, Node.js). The platform features key modules such as live tracking, real-time notifications, user and admin dashboards, booking tools, and integrated payment options—making the entire event planning and execution process more efficient for both organizers and participants.

2. Problem Statement

Successful event organization depends on smooth communication, effective tracking, and proper coordination among all stakeholders. However, many existing event management tools are disjointed, lack real-time capabilities, and often fail to address local needs. This can lead to problems such as miscommunication, missed timelines, and the absence of live updates—causing frustration for both organizers and attendees. To overcome these challenges, there is a clear need for a unified digital solution that provides real-time tracking, robust management tools, and easy accessibility for both users and administrators.

3. Problem Solution for Proposed System

WZH Moments offers a powerful web-based platform that brings every aspect of event management into one place. Users can sign up, explore events, make bookings, and stay informed with real-time updates. Organizers have the tools to create, modify, and manage their events while tracking live progress through interactive dashboards. The admin panel includes features for data analytics, user oversight, and event approval workflows. Built on the MERN stack and enhanced with real-time technologies like Socket.IO, the platform delivers a smooth, interactive, and always up-to-date experience for everyone involved.

4. Related System Analysis/Literature Review

Application Name	Identified Weaknesses	How WZH Moments Addresses It
Eventbrite	Offers only basic tracking and lacks flexibility in customization	Provides full real-time tracking, customizable modules, and a platform tailored to local needs
Cvent	High cost and a complex, less intuitive interface	Delivers a cost-effective solution with a clean, user-friendly design
Local Event Portals	Feature limitations and absence of real-time updates	Includes real-time tracking and a wide range of robust features

Most current platforms fall short when it comes to integrated real-time tracking and customization based on local needs. **WZH Moments** sets itself apart by offering live updates and dedicated user and admin panels specifically designed to cater to the event management landscape in Pakistan.

5. Benefits of Proposed System

- Live tracking of events and task progress for better visibility
- Centralized dashboards for both admins and users to manage activities efficiently
- Seamless booking process with integrated payment options
- Enhanced communication through real-time notifications and chat features
- Insightful analytics to monitor event performance and attendance trends
- User-friendly interface designed to offer a smooth and intuitive experience
- Scalable system architecture, built to support future growth and new features

6. Scope

The first version of **WZH Moments** will focus on delivering a web-based solution for managing both public and private events—such as weddings, seminars, and parties. Key features will

include event creation, booking management, real-time tracking, and user/admin controls. Looking ahead, the platform may expand to include mobile applications, integration with third-party services like caterers and photographers, and AI-driven analytics to help optimize event planning and execution.

7. Modules

7.1 Event Creation & Management

Organizers can easily create and edit events, add detailed schedules, specify venues, and provide event descriptions—all through an intuitive interface.

7.2 Run-Time Tracking

Real-time tracking of event activities and task progress is available for both admins and users, with visual progress updates powered by technologies like Socket.IO.

7.3 Booking & Payments

Users can seamlessly book event tickets or services, complete secure online payments, and receive instant confirmations.

7.4 User Panel

Includes features for managing user profiles, viewing event history, tracking upcoming bookings, and receiving timely notifications.

7.5 Admin Panel

Admins can approve or reject events, access analytics dashboards, manage users, and control all event-related activities from a centralized panel.

7.6 Notifications and Communication

Automated alerts and reminders keep participants informed, while real-time chat supports direct communication between organizers and attendees.

7.7 Analytics and Reporting

Provides detailed reports on each event, including attendance statistics, user feedback, and key performance metrics to help organizers evaluate success.

8. System Limitations & Constraints

- Real-time features require a stable internet connection to function effectively
- Overall system performance is influenced by the quality and capacity of the server infrastructure
- The initial release is limited to web-based access, with no native mobile application available at launch
- User adoption and platform success will heavily rely on effective marketing and outreach efforts

9. Software Process Methodology

The development of **WZH Moments** will follow the Agile (Scrum) methodology, ensuring a flexible and adaptive approach. The project will be divided into manageable sprints, each focusing on specific feature modules. This iterative process allows for continuous integration, rapid prototyping, and regular feedback—leading to faster, more efficient development and delivery of a high-quality product..

10. Tools & Technologies

Tools/Technologies	Version	Rationale
MongoDB	Latest	NoSQL database for scalability
Express.js	Latest	Backend API framework
React.js	Latest	Frontend user interface
Node.js	Latest	Backend server

Tools/Technologies	Version	Rationale
Socket.IO	Latest	Real-time communication
Git/GitHub	Latest	Version control
Figma	Latest	Mockup/designs
VS Code	Latest	Development environment

11. Project Stakeholders and Roles

The following are the stakeholders of our projects.

Project Sponsor	COMSATS University Islamabad, Sahiwal Campus
Stakeholder	• M. Haris Munir • M. Zulal Saleem • Wali Muhammad • Dr. Zafar Iqbal • Man Tehreem • Final Year Project Committee: Evolution of project

12. Team Members Individual Tasks/Work Division

Name	Registration no	Role/Tasks
M. Haris Munir	FA22-BCS-069	Documentation

Name	Registration no	Role/Tasks
		Web Design Front end
M. Zulal Saleem	FA22-BCS-093	Backend Data Gathering
Wali Muhammad	FA22-BCS-121	Data Analysis Model Training

13. Data Gathering Approach

- **Online Surveys:** Conducted to collect user requirements, preferences, and expectations
- **Interviews:** Held with local event organizers to understand real-world challenges and needs
- **Competitor Analysis:** Studied existing event management platforms to identify gaps and improvement areas
- **Mock User Stories:** Created to map out functional flows and ensure the platform meets user needs effectively

14. Concepts

14.1 Real-Time Tracking

Utilizes **Socket.IO** to provide live status updates and real-time communication between users, organizers, and the system.

14.2 Database

MongoDB is used to manage and store key data such as user profiles, event details, bookings, and system logs efficiently.

14.3 Data Analytics

Event-related data is aggregated and analyzed to generate insightful reports and visual dashboards for performance evaluation.

14.4 Deployment

The application is deployed on cloud platforms like **Vercel** or **Heroku**, ensuring scalability and global accessibility.

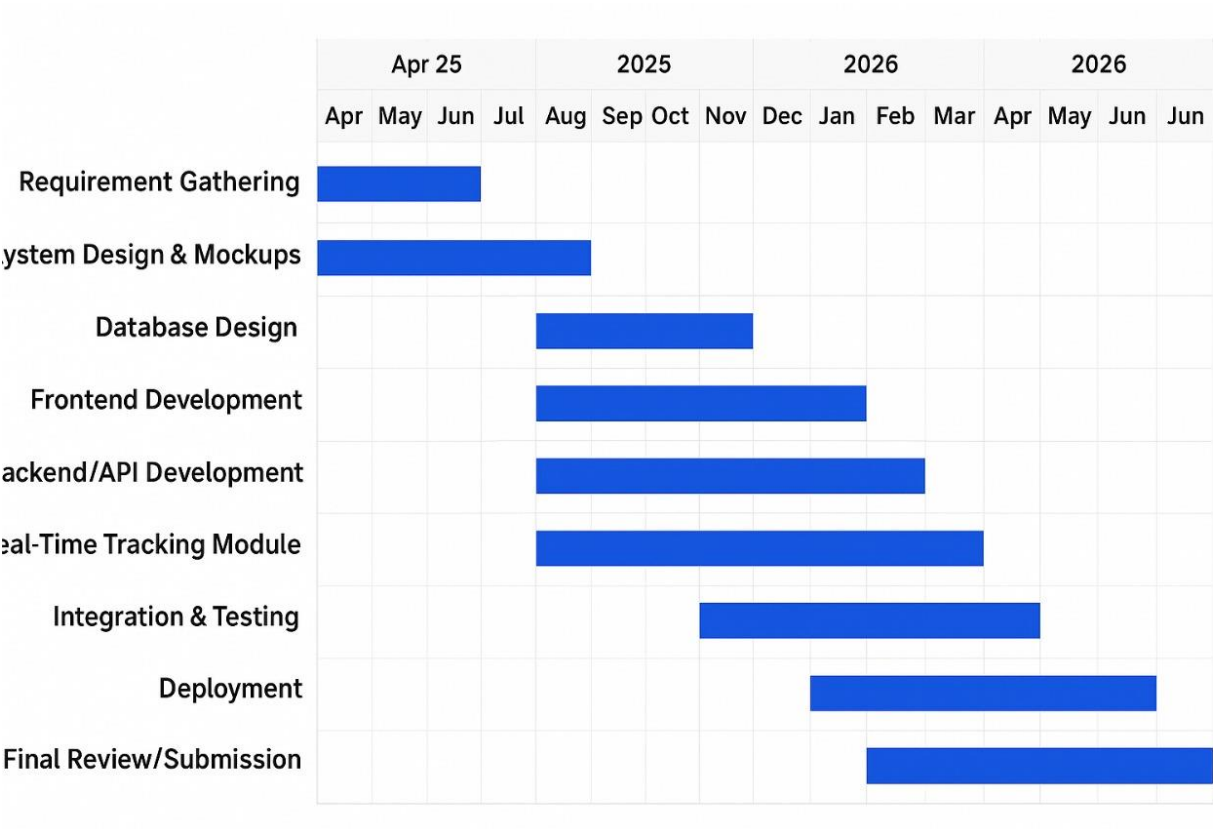
14.5 Frontend and Backend

Built using **React.js** for a responsive user interface and **Node.js/Express** for backend services, following a **RESTful architecture** for efficient API communication.

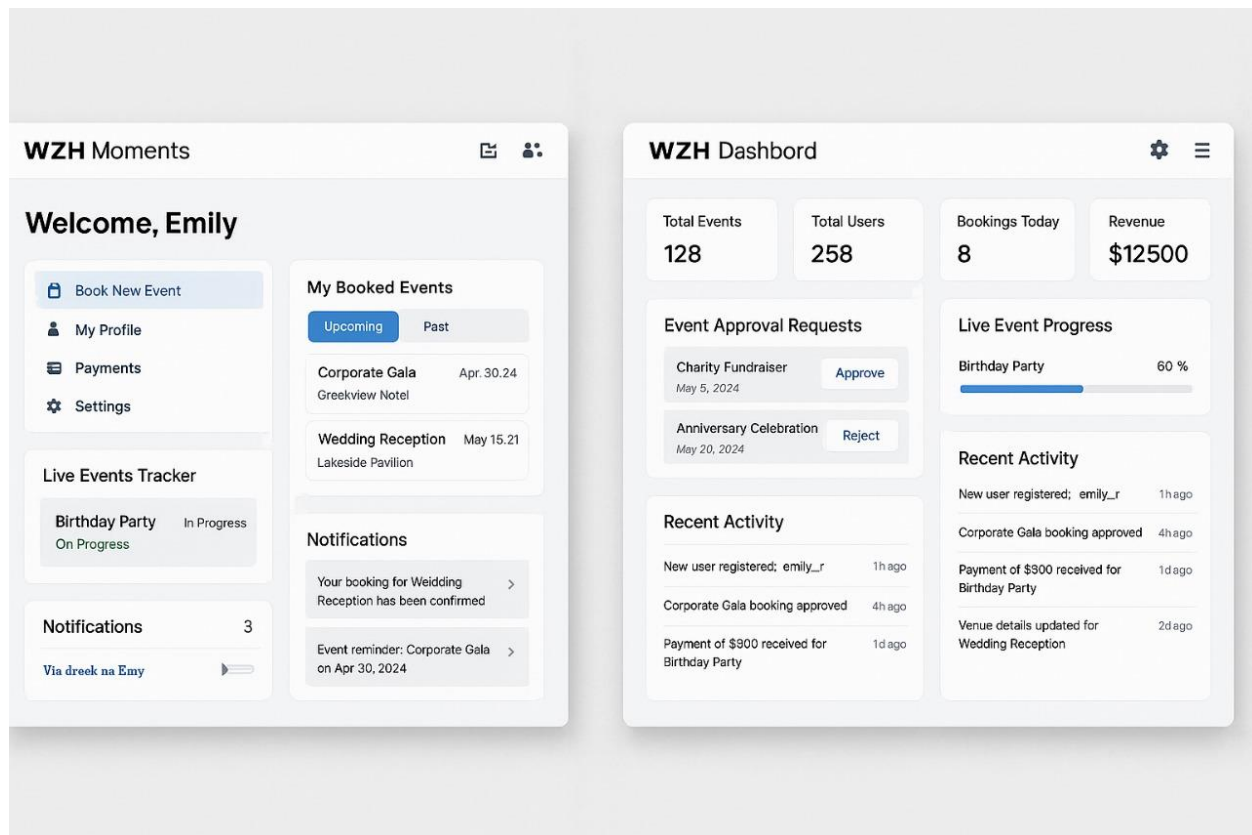
14.6 Version Control

Git and **GitHub** are used to manage source code, enabling collaborative development and version tracking across the team.

15. Gantt Chart



16. Mockups



17. Conclusion

WZH Moments is set to redefine event management by providing a real-time, user-friendly platform built on modern web technologies. By centralizing event creation, tracking, booking, and communication, the platform empowers both organizers and users for successful, hassle-free events.

18. References

1. [Eventbrite Developer Docs](#)
2. [Socket.IO Real-time Docs](#)
3. [MERN Stack Guide](#)